

# Vaso constrictors

LECTURE-5 Local Anesthesia

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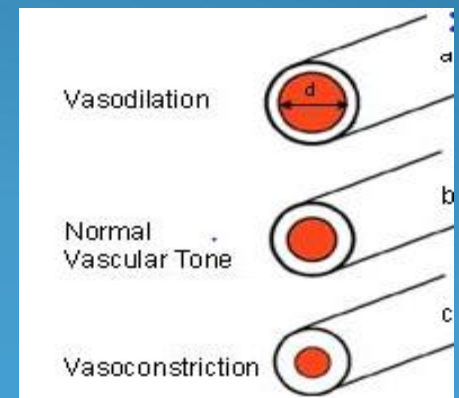
# Course learning outcomes

- List the vasoconstrictors in local anesthetic agents used in dentistry
- Define the pharmacological properties of vasoconstrictors
- Explain need of vasoconstrictors

**These are drugs that constrict blood vessels and thereby control tissue perfusion.**

- 1. By constricting blood vessels, vasoconstrictors decrease blood flow to the site of injection.**
- 2. Absorption of the LA into the bloodstream is slowed, thereby producing lower levels in the blood.**
- 3. Lower anesthetic blood Levels decrease the risk of an overdose reaction.**
- 4. Increasing the duration of most LAs.**
- 5. Decreasing bleeding at the site of surgery.**

## Vasoconstrictors (VC)



## Types of vasoconstrictors:

The vasoconstrictors used in conjunction with LAs are **chemically identical or quite similar to the sympathetic nervous system** mediators epinephrine and norepinephrine and therefore are called sympathomimetic drugs.

### 1. Catecholamines:

- a) Epinephrine.
- b) Nor-epinephrine.
- c) Dopamine.

### 2. Non-catecholamines:

- a) Amphetamine.
- b) Ephedrine.

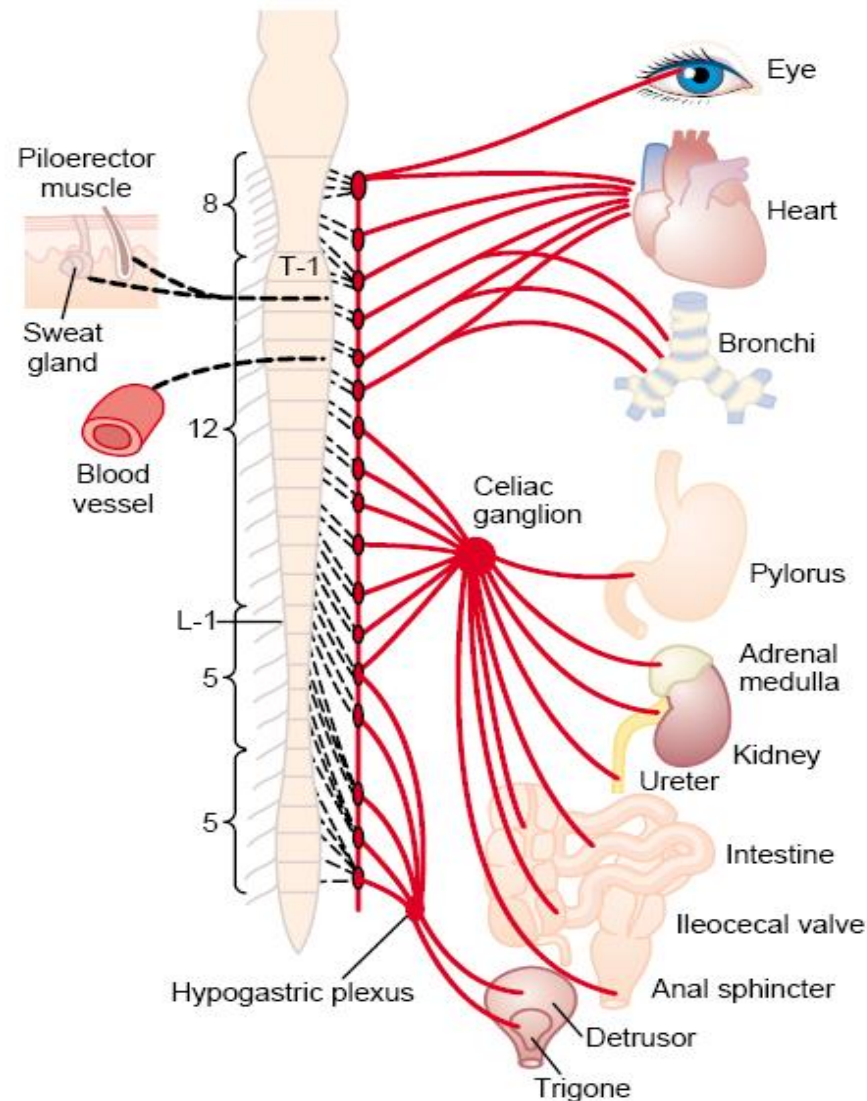
### 3. Felypressin, synthetic analogue of Vaso-pressin (anti-diuretic hormone).



# Fight or Flight



# Sympathetic nervous system



*Autonomic nervous system.*

two major subdivisions called the *sympathetic nervous system* and the *parasympathetic nervous system*

control arterial pressure, gastrointestinal motility, gastrointestinal secretion, urinary bladder emptying, sweating, body temperature, and many other activities,



# Adrenergic Receptors

These are found in most tissues of the body. There are two types of adrenergic receptors:  $\alpha$  &  $\beta$ .

**1. Alpha receptors ( $\alpha$ ):** activation of  $\alpha$  receptors causes contraction of smooth muscle in blood vessels (vasoconstriction)

**2. Beta receptors ( $\beta$ ):** activation of  $\beta$  receptors produces smooth muscle relaxation (vasodilation and bronchodilation) and cardiac stimulation (increased HR and force of contraction).

They are further divided into:

**a)  $\beta_1$ ,** found in the heart and small intestine and are responsible for cardiac stimulation and lipolysis.

**b)  $\beta_2$ ,** found in the bronchi, vascular beds, and uterus and produce bronchodilation and vasodilation.

## Concentrations (dilution) of vasoconstrictors:

The concentration is commonly referred to as a ratio,

e.g., 1:1000 means there is 1 gm (1000 mg) of drug in 1000 ml of solution,

therefore 1:1000 concentration contains **1mg/ml** of solution.

1:10,000

0.1 mg/ml

1:80,000

0.0125 mg/ml

**1:100,000**

**0.01 mg/ml** (used in dentistry)

**1:200,000**

**0.005 mg/ml** (used in dentistry)

- **Epinephrine remains the most effective and most used vasoconstrictor in medicine and dentistry.**
- **Nor-epinephrine is not recommended in dentistry, it produces intense peripheral vasoconstriction.**



**Epinephrine (adrenaline):** acts on both  $\alpha$  and  $\beta$  adrenergic receptors, with its  $\beta$  effects predominate.

**Systemic actions:**

1. **Myocardium**: both cardiac output and heart rate are increased (due to stimulation of  $\beta_1$  receptors).
2. **BP**: there is increased systolic pressure ( $\alpha$  effect).
3. **RS**: adrenaline is a potent dilator ( $\beta_2$  effect) of the smooth muscles of the bronchioles. It is the drug of choice for management of severe asthma (bronchospasm ).
4. **CNS**: its CNS-stimulating actions become prominent when an excessive dose is administered.
5. **Coronary artery**: it produces dilation of the coronary arteries, leading to increased coronary artery blood flow.



The New Arab

**Saudi police bust uncovers millions of amphetamine pills hidden in packets of Mate green tea**

Visit



## 6. CVS: its action is direct stimulation:

Increased systolic and diastolic pressures, increased cardiac output, increased stroke volume, increased HR, increased strength of contraction, increased myocardial oxygen consumption.

### Clinical applications:

1. Management of **acute allergic reactions**.
2. Management of **bronchospasm**.
3. Treatment of **cardiac arrest**.
4. As a vasoconstrictor for **hemostasis**.
5. As a vasoconstrictor in LAs, to **decrease absorption into the systemic circulation & increase duration of action**.
6. To produce mydriasis (dilatation of the pupils).

### Cons. with epinephrine

### LA (generic)

1:50,000

Lidocaine

1:80,000

Lidocaine (UK only)

1:100,000

Lidocaine (USA,CA)

1:200,000

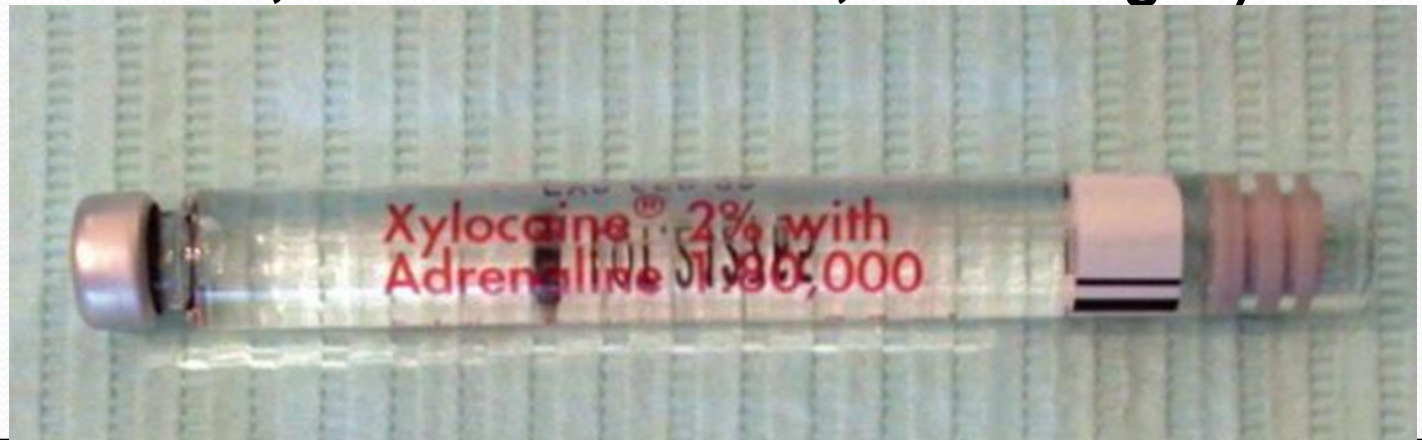
Bupivacaine

Shelf life of LAs containing vasoconstrictors is shorter Because vasoconstrictors gets oxidized .

### Maximum dose of Vasoconstrictors

Normal healthy pt: **0.2 mg per appointment** (20 ml of 1:100,000 concentration, 11 cartridges).

Pt with clinically significant cardiovascular impairment: **0.04 mg / appointment** (4 ml of 1:100,000 concentration, 2 cartridges).





# **Nor-epinephrine: not recommended in dentistry**

Extravascular injections of Norephrinpherine may produce necrosis and sloughing of tissues because of **intense  $\propto$  stimulation**



Palatal mucosal necrosis after  
administration of a palatal infiltration

# Contraindications for the use of vasoconstrictors:

1. Patients with **high BP.**
2. Patients with **cardiovascular disease.**
3. Patients with **hyperthyroidism.**
4. Patients receiving **MAO inhibitor and tricyclic antidepressants.**

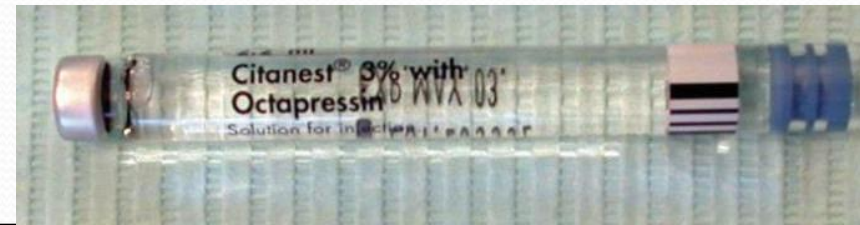


**Felypressin (octapressin)**: it is a synthetic analogue of the antidiuretic hormone vasopressin.

1. Non-sympathomimetic amine.
2. Its action appears to be more pronounced on the venous than on the arteriolar microcirculation.
3. No effect on adrenergic nerve transmission, so safely given to:
  - a) Hyperthyroid patients.
  - b) Patients on MAO inhibitors.
  - c) Tricyclic antidepressants.
4. **Contraindicated for pregnant women.**

It is structurally similar to Oxytocin, Can cause uterine contractions.

**Availability:** It is employed in a concentration of 0.03 I.U. /ml with 3% prilocaine .





Thank you